

Wave-Optics Gravitational Lensing of Gravitational Waves
in a Hierarchical Triple: a Worked Derivation

Final project — Gravitational Waves and AI-Assisted Research

September 2026

Abstract

This chapter derives, from the linearized wave equation, the formalism used to compute how a gravitational wave (GW) from a compact binary is lensed by a third, more massive body in the same hierarchical triple. It follows the scalar-field reduction of GW propagation in a weak-field lens background through to the Kirchhoff–Fresnel diffraction integral, works out the point-mass lens in closed form and cross-checks it against two independent numerical methods, derives the specific geometry of a source and lens that share one physical system (as opposed to the usual well-separated cosmological lens), and closes with the two regimes — a static lens during a fast chirp, and a moving lens sampled by a slowly-evolving quasi-monochromatic source — that `cases/case_A_chirp` and `cases/case_B_monochromatic` instantiate numerically. Every numbered check below is implemented and run in `tests/`; this document derives *why* each check is the right one, not just that it passes.

Contents

1	Setup and scope	2
1.1	The physical picture	2
2	From the wave equation to a scalar field	4
2.1	The wave equation on a weak-field background	4
2.2	The Helmholtz equation and the thin lens	5
3	The point-mass lens	6
3.1	Geometric optics: images, magnification, time delay	6
3.2	The exact result: a confluent hypergeometric function	7
3.3	A fast hybrid evaluator	8
4	The hierarchical triple as a lens system	9
4.1	The outer orbit	9
4.2	D_{LS} is not what it is in a textbook lens	9
4.3	Where the thin-lens/paraxial approximation itself is being pushed	10
5	The inner binary as a source	12
6	Putting it together: two regimes	13
6.1	Static lens (Case A)	13
6.2	Quasi-monochromatic, moving lens (Case B)	13
6.3	An idealized detector	13
	References	15

Chapter 1

Setup and scope

1.1 The physical picture

The system is a hierarchical triple: an inner compact binary of masses m_1, m_2 (the GW *source*) orbits, at semi-major axis a_{in} , a common centre of mass which itself orbits a third, more massive body of mass M_L (the *lens*) at semi-major axis a_{out} , with

$$a_{\text{in}} \ll a_{\text{out}}. \quad (1.1)$$

This is the ordinary stability condition for a hierarchical triple, and it does double duty here: it is also exactly the condition under which the inner binary can be treated, for lensing purposes, as a single point source — both members are deflected identically, so there is no differential lensing across the binary itself (no "binary microlensing" substructure). Throughout, $M_L \gg m_1 + m_2$, so the lens sits, to the accuracy used here, at rest at the triple's barycentre while the inner binary (as a whole) orbits it.

The triple is taken to be within our own Galaxy (or a similarly nearby system), so the lens and source are at essentially the same distance from Earth and cosmological redshift plays no role: distances D_L (observer to lens), D_S (observer to source) and D_{LS} (lens to source) are ordinary Euclidean distances, and the lens redshift is $z_L = 0$ (the factor $(1 + z_L)$ that multiplies the lens mass everywhere in the cosmological lensing literature is simply 1 here). Chapter 4 works out a consequence of this that a textbook cosmological treatment does not have to face: D_{LS} is *not* a large, slowly-varying distance — it is set by the outer orbit itself, and can be small, zero, or even formally negative.

Two epochs of the same inspiral

The inner binary loses energy to gravitational radiation and inspirals. Two snapshots of that inspiral are studied:

Case A — the chirp. Late inspiral: the GW frequency sweeps through the sensitive band of a ground-based-like detector in seconds. That is enormously faster than any plausible outer orbital period, so the lens is *frozen* at a single impact parameter for the whole observation.

Case B — quasi-monochromatic. Early inspiral: the inner binary is wide, the GW frequency is low and drifts negligibly over an observation spanning several outer periods. Now the outer orbital motion *is* resolved, and the impact parameter sweeps periodically.

Both regimes are made precise, and checked against the actual numbers used in **cases/**, in Chapter 6.

Choice made

The lens mass, $M_L \sim 5 \times 10^4 M_\odot$ (an “IMBH”, between the conventional intermediate-mass and supermassive black hole ranges), is chosen so that the wave-optics parameter w (defined in Eq. 2.6) is not astronomically extreme at either epoch’s frequency, not because it is a typical hierarchical-triple lens mass in the literature. `wiki/log.md` records this choice and the numbers it produces; it is not a fitted or tuned value beyond that one requirement.

Chapter 2

From the wave equation to a scalar field

The user-facing goal of this project stops here: reaching the point where gravitational-wave lensing is a *scalar*-field diffraction problem, and justifying why that is legitimate.

2.1 The wave equation on a weak-field background

Write the full metric as a background plus a small perturbation, $g_{\mu\nu} = \bar{g}_{\mu\nu} + h_{\mu\nu}$, where $\bar{g}_{\mu\nu}$ is the lens's own weak-field metric (Newtonian potential $U = -GM_L/r$, $|U|/c^2 \ll 1$ everywhere relevant here since the lens is far outside its own Schwarzschild radius at the impact parameters considered) and $h_{\mu\nu}$ is the GW. In Lorenz gauge on this background, the linearized vacuum Einstein equations give a wave equation for $h_{\mu\nu}$ with a curvature-coupling term,

$$\bar{\square} h_{\mu\nu} - 2\bar{R}_{\mu\alpha\nu\beta} h^{\alpha\beta} = 0. \quad (2.1)$$

This is the standard starting point for GW propagation on a curved background (e.g. Isaacson, 1968). The high-frequency (eikonal) regime relevant here is set by comparing the GW wavelength λ_{GW} to the background's curvature length scale $\mathcal{R}$: the ray/thin-lens treatment below is valid when $\lambda_{GW} \ll \mathcal{R}$. For a point lens $\mathcal{R} \sim R_E$ (the Einstein radius), and the curvature-coupling term in Eq. (2.1) is then subdominant to $\bar{\square} h_{\mu\nu}$ by a factor $\sim (\lambda_{GW}/\mathcal{R})^2$ (Peters, 1974; Takahashi and Nakamura, 2003, §2). Dropping it, each independent polarization amplitude $h_{\mu\nu} = e_{\mu\nu} \Phi$ (fixed polarization tensor $e_{\mu\nu}$, transverse-traceless) obeys the *same* equation as a massless scalar field on the lens background,

$$\bar{\square} \Phi = 0. \quad (2.2)$$

This is the key reduction: lensing in this regime does not mix or rotate the two GW polarizations, and every polarization is amplified/retarded by the identical (complex, frequency-dependent) factor derived below. It is also why solving the scalar-field diffraction problem *is* solving the GW lensing problem here, and nothing about spin-2 structure survives past this point.

Check

This reduction is only used, never re-derived numerically in this project — it is a standard result cited above. What *is* checked numerically, repeatedly below, is everything built on top of it: the diffraction integral it leads to, the point-lens closed form, and its geometric-optics limit.

2.2 The Helmholtz equation and the thin lens

For the weak-field metric $\bar{g}_{\mu\nu} = \eta_{\mu\nu} + 2U\delta_{\mu\nu}$ (isotropic weak-field form, $c = 1$ units in this section), Eq. (2.2) for a wave of fixed angular frequency ω , $\Phi(\mathbf{r}, t) = \psi(\mathbf{r})e^{-i\omega t}$, becomes

$$(\nabla^2 + \omega^2)\psi = 4\omega^2 U\psi, \quad (2.3)$$

a Helmholtz equation with the lens potential acting as a (frequency -dependent) refractive-index perturbation, $n(\mathbf{r}) = 1 - 2U(\mathbf{r})$ — exactly analogous to geometric/wave optics in a medium of varying refractive index, which is the origin of calling this whole formalism “wave optics.”

Thin-lens (Fresnel–Kirchhoff) reduction

Because $a_{\text{out}} \ll D_L, D_S$ (Eq. 1.1 and the Galactic -scale distances above), the deflection happens in a region whose extent along the line of sight is negligible compared to D_L and D_S : the standard thin-lens approximation. Propagating Eq. (2.3) from the source plane to the lens plane and on to the observer via the Fresnel diffraction integral (paraxial approximation, valid since all angles involved are $\ll 1$) gives the amplification factor — the ratio of the lensed to unlensed wave at the observer,

$$F(f) = \frac{D_S}{D_L D_{LS}} \frac{f}{i} \int d^2\xi \exp[i2\pi f t_d(\xi; \eta)], \quad (2.4)$$

where ξ is position on the lens plane, η the (fixed) source position, and t_d the time delay (geometric + Shapiro) of a ray through ξ relative to the undeflected path (Schneider et al., 1992; Takahashi and Nakamura, 2003). By construction, $h_{\text{lensed}}(f) = F(f) h_{\text{unlensed}}(f)$: everything the lens does to the (scalar-equivalent) waveform is contained in this single complex, frequency-dependent number.

Dimensionless form

Scaling the lens-plane position by the Einstein radius $\xi_0 \equiv R_E$ and the source position by the corresponding source-plane scale $\eta_0 \equiv \xi_0 D_S / D_L$, with $\mathbf{x} \equiv \xi / \xi_0$, $\mathbf{y} \equiv \eta / \eta_0$ (so $y = |\mathbf{y}|$ is the dimensionless impact parameter used throughout this project), and defining the dimensionless deflection potential $\psi(\mathbf{x})$ via $t_d = (1 + z_L) \frac{D_S}{D_L D_{LS}} [\frac{1}{2}|\mathbf{x} - \mathbf{y}|^2 - \psi(\mathbf{x})]$, Eq. (2.4) becomes

$$F(w, \mathbf{y}) = \frac{w}{2\pi i} \int d^2\mathbf{x} \exp[iw\varphi(\mathbf{x}, \mathbf{y})], \quad \varphi(\mathbf{x}, \mathbf{y}) \equiv \frac{1}{2}|\mathbf{x} - \mathbf{y}|^2 - \psi(\mathbf{x}), \quad (2.5)$$

where the dimensionless frequency is

$$w \equiv 8\pi \frac{GM_L}{c^3} f \quad (2.6)$$

(the $(1 + z_L)$ factor is 1 here, per Chapter 1). This is exactly the function `F_bruteforce_2d` in `waveoptics.py`: no closed form has been used yet, only Eq. (2.3) and the thin-lens/paraxial reduction.

Physically, a wave much longer than the lens’s effect scale should not be lensed at all: $F(w, \mathbf{y}) \rightarrow 1$ as $w \rightarrow 0$. This is not obvious from Eq. (2.5) by naive Taylor expansion (the integral is only conditionally convergent, §3), but it is confirmed numerically to $\sim 5 \times 10^{-4}$ at $w = 10^{-4}$ by `tests/test_waveoptics.py::check_low_w_limit`.

Chapter 3

The point-mass lens

Everything above is general (any $\psi(\mathbf{x})$). From here on the lens is a single point mass, $\psi(\mathbf{x}) = \ln|\mathbf{x}|$, the deflection potential whose gradient $\nabla\psi = \mathbf{x}/|\mathbf{x}|^2$ gives the standard $1/|\mathbf{x}|$ deflection law of a point mass.

3.1 Geometric optics: images, magnification, time delay

By symmetry take $\mathbf{y} = (y, 0)$. Stationary points of $\varphi(\mathbf{x}, \mathbf{y})$ (Fermat’s principle: images sit where the time delay is stationary) restricted to the $x_2 = 0$ axis solve

$$x - \frac{1}{x} = y \implies x_{\pm} = \frac{1}{2} \left(y \pm \sqrt{y^2 + 4} \right), \quad x_+ x_- = -1. \quad (3.1)$$

Two images, always: $x_+ > 1 > 0 > x_- > -1$ for $y > 0$.

Magnification

For an axisymmetric lens the 2D lens map $\mathbf{y}(\mathbf{x}) = f(x)\hat{\mathbf{x}}$, $f(x) \equiv x - 1/x$, has Jacobian $\det(\partial\mathbf{y}/\partial\mathbf{x}) = (f(x)/x) f'(x)$ (the product of the tangential and radial stretch factors — a standard identity for axisymmetric maps, e.g. Schneider et al., 1992). With $f'(x) = 1 + 1/x^2$,

$$\mu(x) = \frac{1}{\det(\partial\mathbf{y}/\partial\mathbf{x})} = \frac{1}{\left(1 - \frac{1}{x^2}\right) \left(1 + \frac{1}{x^2}\right)} = \frac{x^4}{x^4 - 1}. \quad (3.2)$$

Check total magnification vs. Paczynski (1986)

The total (unsigned) magnification $\mu_{\text{tot}}(y) = |\mu(x_+)| + |\mu(x_-)|$ built from Eqs. (3.1)–(3.2) must equal the independent, textbook microlensing formula $A(y) = (y^2 + 2)/\left(y\sqrt{y^2 + 4}\right)$ (Paczynski, 1986). Verified to floating-point precision ($< 3 \times 10^{-15}$, all y tested) in `tests/test_waveoptics.py::check_paczynski_magnification` — a genuine cross-check, since Eq. (3.2) was derived here from the Jacobian, not copied from Paczyński’s paper.

Time delay and the geometric-optics amplification factor

The (dimensionless) time delay of each image is $\varphi(x_{\pm}, y)$, and the delay of the (always positive-parity, “minimum”) image x_+ relative to itself is trivially zero, while the saddle-point image x_- arrives later by $\Delta T(y) \equiv \varphi(x_-, y) - \varphi(x_+, y) > 0$. Evaluating Eq. (2.5) by stationary

phase (valid as $w \rightarrow \infty$) sums the two images with Morse-index phases (minimum: 0; saddle: $-\pi/2$, i.e. an overall factor of $-i$, in the $h(f) = \int h(t)e^{-i2\pi ft}dt$ convention fixed in `wiki/conventions.md`) and each image's own, un-subtracted phase $w\varphi(x_+, y)$:

$$F_{\text{geo}}(w, y) = e^{iw\varphi(x_+, y)} \left[\sqrt{\mu(x_+)} - i\sqrt{|\mu(x_-)|} e^{iw\Delta T(y)} \right]. \quad (3.3)$$

An earlier version of Eq. (3.3) omitted the leading $e^{iw\varphi(x_+, y)}$ factor (implicitly assuming a reference phase subtraction that Eq. (2.5) does not make). This gave the right $|F|$ but the wrong phase compared to the exact closed form (§3.2) at large w . The error was diagnosed by comparing the two directly and recognizing the residual phase scaled linearly with w at the rate $\varphi(x_+, y)$ — and fixed by restoring the missing factor. After the fix, `tests/test_waveoptics.py::check_geometric_optics_limit` finds relative agreement 3×10^{-4} at $w = 10$, shrinking to 3×10^{-5} at $w = 1000$, the expected behaviour of an asymptotic ($w \rightarrow \infty$) expansion. See `wiki/log.md` for the full account.

3.2 The exact result: a confluent hypergeometric function

Solving Eq. (2.5) for the point-mass lens exactly (originally Peters, 1974; Deguchi and Watson, 1986; the standard form quoted in the GW-lensing literature is Takahashi and Nakamura, 2003) gives

$$F(w, y) = \exp\left[\frac{\pi w}{4} + i\frac{w}{2} \ln \frac{w}{2}\right] \Gamma\left(1 - \frac{iw}{2}\right) {}_1F_1\left(\frac{iw}{2}, 1; \frac{iwy^2}{2}\right). \quad (3.4)$$

This is `src/gwlens/waveoptics.py::F_point_lens`, evaluated with arbitrary-precision complex arithmetic (`mpmath`, since `scipy.special.hyp1f1` does not accept complex parameters). Rather than trust Eq. (3.4) on citation alone, it is checked against two independently-derived numerical evaluations of Eq. (2.5) itself.

Independent check 1: the 1D Bessel-reduced integral

Writing $|\mathbf{x} - \mathbf{y}|^2 = x^2 + y^2 - 2xy \cos \theta$ in polar coordinates on the lens plane and using the standard identity $\int_0^{2\pi} d\theta e^{ia \cos \theta} = 2\pi J_0(a)$ (Abramowitz & Stegun 9.1.21) to do the angular integral in Eq. (2.5) analytically leaves a 1D radial integral,

$$F(w, y) = -i w e^{iwy^2/2} \int_0^\infty dx x J_0(wxy) \exp\left[iw\left(\frac{1}{2}x^2 - \ln x\right)\right]. \quad (3.5)$$

This integral is only conditionally convergent; it is regularized with the standard $i\epsilon$ prescription (Ulmer and Goodman, 1995), $w \rightarrow w(1 + i\eta)$, i.e. a convergence factor $e^{-\eta wx^2/2}$, evaluated for a sequence of $\eta \rightarrow 0$ and extrapolated. `src/gwlens/waveoptics.py::F_radial_1d` implements this, with the cutoff radius and the number of quadrature sub-intervals set automatically from η and w (so as to resolve the local oscillation period rather than a hand-tuned grid).

At $w = 1, y = 1$: relative error vs. Eq. (3.4) is 2.6×10^{-2} at $\eta = 0.02$, 1.3×10^{-2} at $\eta = 0.01$, 6.5×10^{-3} at $\eta = 0.005$ — each halving of η almost exactly halves the error (ratios 1.98, 1.99), the expected $O(\eta)$ scaling of the regularization itself. `tests/test_waveoptics.py::check_radial_1d_converges_to_closed_form`.

Independent check 2: brute-force 2D quadrature

Eq. (2.5) evaluated directly by 2D quadrature over a regularized (Gaussian-damped) square grid, with no angular reduction and no closed form anywhere — the least clever, most first-principles evaluator in the project. At $w = 1$, $y = 1$, $x_{\max} = 25$, 1200×1200 grid points, $\eta = 0.005$: relative error 1.1×10^{-2} , consistent with the same $O(\eta)$ regularization error seen in the 1D check. `tests/test_waveoptics.py::check_bruteforce_2d_agrees`; `waveoptics.py::F_bruteforce_2d`.

check_einstein_ring_diffraction_spike

At exact alignment ($y = 0$) geometric optics diverges ($\mu(x_+=1)$ is singular, Eq. 3.2) — the textbook point-source caustic. The exact $F(w, 0)$ from Eq. (3.4) stays finite for every w tested (1.81, 3.96, 7.93 at $w = 1, 5, 20$): diffraction regularizes the caustic, the GW analogue of the optical Arago/Poisson spot (Deguchi and Watson, 1986). `tests/test_waveoptics.py::check_einstein_ring_regularization`.

3.3 A fast hybrid evaluator

Eq. (3.4)’s confluent hypergeometric series converges slowly once its argument $iwy^2/2$ is large — exactly the regime, large w , where Eq. (3.3) is independently validated to agree with it to better than 0.4%. `waveoptics.py::F_hybrid` therefore uses Eq. (3.4) below $w = 30$ and Eq. (3.3) above it (the threshold checked directly, `tests/test_waveoptics.py::check_hybrid_matches_at_threshold`: 3.5% agreement at $w = 30$ across several y). This was not an optimization made in advance: evaluating Eq. (3.4) directly across the frequency sweep actually used in Case A took ~ 150 s; `F_hybrid` needs none of that (see `wiki/log.md`).

Chapter 4

The hierarchical triple as a lens system

4.1 The outer orbit

The source’s position relative to the (effectively fixed) lens follows an ordinary Kepler orbit: elements $(a_{\text{out}}, e_{\text{out}}, i_{\text{out}}, \Omega_{\text{out}}, \omega_{\text{out}})$, solved by Newton’s method on Kepler’s equation $M = E - e \sin E$ and rotated into the observer frame (z along the line of sight) by the standard sequence $R_z(\Omega)R_x(i)R_z(\omega)$ (e.g. Murray and Dermott, 1999):

$$x_{\text{sky}} = r [\cos \Omega \cos \theta - \sin \Omega \sin \theta \cos i], \quad (4.1)$$

$$y_{\text{sky}} = r [\sin \Omega \cos \theta + \cos \Omega \sin \theta \cos i], \quad (4.2)$$

$$z_{\text{los}} = r \sin \theta \sin i, \quad \theta \equiv \nu + \omega, \quad (4.3)$$

r, ν the orbital radius and true anomaly.

Changelog

Kepler’s equation solver inverts to machine precision for $e \in \{0, 0.3, 0.7, 0.95\}$; a circular orbit gives exactly constant r ; an edge-on orbit ($i = \pi/2, \Omega = \omega = 0$) projects to a line ($y_{\text{sky}} \equiv 0$ to 6×10^{-17}); a face-on orbit ($i = 0$) reproduces the orbital-plane position exactly. All in `tests/test_geometry.py`.

4.2 D_{LS} is not what it is in a textbook lens

This is the one place this project’s geometry genuinely differs from the cosmological lensing this formalism was built for, and it is easy to get wrong by carrying over the textbook intuition unexamined. In the usual setting, lens and source are at very different, essentially fixed distances and D_{LS} barely varies. Here they are in the *same* system: the lens sits at D_L , and the source’s distance is $D_S(t) = D_L + z_{\text{los}}(t)$, with $z_{\text{los}}(t)$ from Eq. (4.3) at the AU-to-sub-pc scale of the outer orbit. Since $|z_{\text{los}}|/D_L \sim 10^{-8}$ for the parameters used here, $D_S \approx D_L$ is an excellent approximation *everywhere it appears except in*

$$D_{LS}(t) = D_S(t) - D_L = z_{\text{los}}(t) \quad (4.4)$$

itself, where it is the whole story. The angular Einstein radius,

$$\theta_E(t)^2 = \frac{4GM_L}{c^2} \frac{D_{LS}(t)}{D_L D_S(t)} \approx \frac{4GM_L}{c^2} \frac{z_{\text{los}}(t)}{D_L^2}, \quad (4.5)$$

is therefore itself time-dependent, and **undefined whenever** $z_{\text{los}}(t) \leq 0$ — the source is instantaneously nearer Earth than the lens, so the lens is not between the observer and the source at all, and there is no lensing geometry for that half of the orbit (not an approximation artefact: $F = 1$ there by construction). As $z_{\text{los}}(t) \rightarrow 0^+$, $\theta_E \rightarrow 0$ and $y = \theta/\theta_E(t) \rightarrow \infty$ for any nonzero transverse offset θ : the formalism turns lensing off smoothly and on its own at the front/back crossing, with no special-casing needed beyond excluding $z_{\text{los}} \leq 0$.

This is exactly the origin of *repeated* lensing in a hierarchical triple: only the far-side half of each outer orbit is lensed at all, so a periodically-orbiting source produces periodic lensing pulses, once per outer period, rather than continuous modulation (D’Orazio and Loeb, 2020, in the geometric-optics limit; the same structure appears here in wave optics, Case B, §6.2).

For the (near edge-on, $i_{\text{out}} = 87^\circ$) orbit used in `cases/`, almost exactly half the orbit is lensed (49.7–50.0% across independent samplings), and y is finite and positive throughout the lensed half. `tests/test_geometry.py::check_time_varying_impact_parameter`.

4.3 Where the thin-lens/paraxial approximation itself is being pushed

§2 derived Eq. (2.4) under a paraxial (small-angle) Fresnel approximation, which additionally requires the physical Einstein radius on the lens plane, ξ_0 , to be much smaller than the propagation distance D_{LS} itself — an assumption that is automatic in cosmological lensing (where D_{LS} is a fixed, large distance) but is exactly what Eq. (4.5) shows is *not* automatic here, since $D_{LS}(t)$ shrinks to zero twice per outer orbit. Using $\xi_0 = \sqrt{4GM_L D_{LS}/c^2}$ (the point-lens Einstein radius written with $D_S \approx D_L$, §4), the paraxial condition $D_{LS} \gg \xi_0$ reduces to

$$D_{LS}(t) \gg \frac{4GM_L}{c^2} = 2R_{\text{Sch}}(M_L), \quad (4.6)$$

twice the lens’s own Schwarzschild radius — the *same* weak-field condition already invoked in §2 to drop the curvature -coupling term in Eq. (2.1), now applied to the line-of-sight separation instead of the impact parameter. This is a genuine, if narrow, respect in which this project’s geometry pushes past where the standard formalism was built to operate, and it is worth being precise about exactly how much it matters, rather than either ignoring it or re-deriving non-paraxial diffraction (a substantially harder problem, evaluated below as not worth it here).

quantity	value (this triple)
$R_{\text{Sch}}(M_L)$	9.87×10^{-4} AU
D_{LS} at the deepest-lensing point ($y = y_{\text{min}}$)	1.81 AU
$D_{LS}/(2R_{\text{Sch}})$ there	919
ξ_0/D_{LS} there (direct paraxial-angle check)	0.033
time per crossing spent with $D_{LS} \lesssim 2R_{\text{Sch}}$	60 s
that time as a fraction of P_{out}	1.7×10^{-4}

Table 4.1: The paraxial condition, Eq. (4.6), evaluated for the system in `src/gwlens/system.py` by `theory/paraxial_validity.py` (`theory/paraxial_validity_numbers.json`).

Table 4.1 is the answer: at the orbital phase that actually produces the interesting physics in Case B — the point of closest projected approach, $y = y_{\text{min}}$, where the lensing pulses

in `cases/case_B_monochromatic/caseB_repeated_pulses.png` peak — D_{LS} is close to its own *maximum* ($\sim a_{\text{out}} \sin i_{\text{out}}$, since minimum transverse offset and maximum line-of-sight separation occur near the same orbital phase for this geometry), not its minimum, and exceeds $2R_{\text{Sch}}$ by three orders of magnitude. The paraxial approximation only becomes marginal in a window of order 60 seconds around each front/back crossing – a fraction $\sim 1.7 \times 10^{-4}$ of the 4-day outer period – and that window is *already* the region where $y \rightarrow \infty$ and $F \rightarrow 1$ by construction (§4), not a region this project draws any conclusion from. A genuine non-paraxial (wide-angle) diffraction treatment near the crossing would be needed to say anything quantitative about that 60-second window itself, and would require abandoning the Fresnel integral, Eq. (2.4), for a substantially more involved formulation – a well-posed but research-level extension, not attempted here. Given that it would only refine an already-negligible, already-correctly-approximated ($F \approx 1$) sliver of the signal, the added complexity is judged not to be worth it for this project’s scope: this is a stated limitation, checked and quantified rather than an unexamined one.

Chapter 5

The inner binary as a source

The unlensed waveform uses the leading-order (Newtonian/quadrupole, restricted) inspiral: not full post-Newtonian, since the physics under test here is the lensing, not waveform accuracy (a stated scope choice, not an oversight). The frequency evolves as

$$\dot{f} = \frac{96}{5}\pi^{8/3} \left(\frac{G\mathcal{M}}{c^3}\right)^{5/3} f^{11/3}, \quad \mathcal{M} \equiv \frac{(m_1 m_2)^{3/5}}{(m_1 + m_2)^{1/5}}, \quad (5.1)$$

the standard quadrupole-formula chirp rate (Peters, 1964; Peters and Mathews, 1963; Maggiore, 2008). Integrating Eq. (5.1) directly gives the time to (formal) coalescence and the frequency track,

$$\tau(f) = \frac{5}{256}(\pi f)^{-8/3} \left(\frac{G\mathcal{M}}{c^3}\right)^{-5/3}, \quad f(t) = \frac{1}{\pi} \left[\frac{5}{256} \left(\frac{G\mathcal{M}}{c^3}\right)^{-5/3} \right]^{-3/8} (t_c - t)^{-3/8}. \quad (5.2)$$

Check, and a bug caught by it

$f(t=0)$ from Eq. (5.2) must reproduce the f_0 used to define $t_c = \tau(f_0)$ (self-consistency of the inversion) and must agree with direct numerical integration of Eq. (5.1) (`scipy.integrate.solve_ivp`). An early hand-inversion of $\tau(f)$ for $f(t)$ mis-placed an exponent (a bracket-power slip) and failed the first check immediately; after fixing it, agreement with the ODE integration is 6×10^{-9} . `tests/test_chirp.py`.

The (restricted, leading-order) time-domain strain amplitude,

$$h(t) = \frac{4}{D_{\text{eff}}} \left(\frac{G\mathcal{M}}{c^2}\right)^{5/3} \left(\frac{\pi f_{\text{GW}}(t)}{c}\right)^{2/3}, \quad (5.3)$$

increases with frequency, as a real inspiral must. An earlier version of Case A's `run.py` used the frequency-*domain* stationary-phase-approximation scaling $|h(f)| \propto f^{-7/6}$ (which decreases) as if it were this time-domain envelope — caught by eye (the plotted chirp amplitude was shrinking toward merger) and fixed; both scalings are now separate, clearly labelled functions (`restricted_pn_amplitude_td` vs. `fd`) with a regression check in `test_chirp.py` that Eq. (5.3) increases with f . No absolute strain calibration (inclination/polarization prefactors, folded into the leading 4) is claimed anywhere in this project — only relative amplitudes and the lensing amplification factor matter for the physics studied here.

Chapter 6

Putting it together: two regimes

6.1 Static lens (Case A)

The lens is frozen for the whole chirp when the merger duration is much shorter than the outer period, $\tau(f_{\text{start}}) \ll P_{\text{out}}$. For the parameters in `src/gwlens/system.py` ($f_{\text{start}} = 10$ Hz, $P_{\text{out}} = 4$ d): $\tau = 15.2$ s, a fraction 4.4×10^{-5} of P_{out} — static to extremely high accuracy (`tests/test_system.py::check_static_lens_regime`). Lensing is then a single frequency-domain multiplication, $h_{\text{lensed}}(f) = F(f, y_A) h_{\text{unlensed}}(f)$, applied via FFT/IFFT in `cases/case_A_chirp/run.py`.

6.2 Quasi-monochromatic, moving lens (Case B)

Conversely, the lens genuinely moves when the observation spans several outer periods while the GW frequency itself barely evolves: $\tau(f_B) \gg T_{\text{obs}} \gg P_{\text{out}}$. For $f_B = 0.05$ Hz, $T_{\text{obs}} = 6P_{\text{out}}$ (24 d): $\tau = 240.7$ d (60.2 outer periods remain), and the fractional frequency drift over T_{obs} is 4.0% (`tests/test_system.py::check_quasi_monochromatic_regime`).

Why instantaneous (adiabatic) $F(w_B, y(t))$ is the right prescription

Eq. (2.4)’s frequency-domain multiplication assumes a *fixed* lens configuration; it does not directly apply when y itself is a function of time. The separation of scales here is enormous: $y(t)$ varies on the outer-orbital timescale ($P_{\text{out}} \sim$ days), while the GW itself oscillates on a timescale $1/f_B \sim 20$ s — nine orders of magnitude faster. To the wave, the lens configuration is, to superb approximation, constant over any one GW cycle (and over very many consecutive cycles). This is the standard adiabatic/quasi-static approximation: at each instant, apply the amplification factor the lens *would* produce for a static source at that instant’s $y(t)$,

$$z_{\text{lensed}}(t) = F(w_B, y(t)) z_{\text{unlensed}}(t), \quad z(t) \equiv h(t) e^{i\phi(t)} \text{ (analytic signal)}, \quad (6.1)$$

used in `cases/case_B_monochromatic/run.py`. This is the one place this project’s two cases use genuinely different numerical machinery (FFT convolution for Case A, direct time-domain multiplication for Case B), and the reason is this timescale argument, not a difference in the underlying physics.

6.3 An idealized detector

Both cases stop at an idealized, illustrative detector view: the analytic-signal (Hilbert-transform) envelope of $h_{\text{lensed}}(t)$ vs. $h_{\text{unlensed}}(t)$, with no noise curve, no antenna pattern, and no matched filtering. This is a deliberate scope choice (`wiki/conventions.md`): the point of

this project is the wave-optics lensing physics and its visualization, not a detection-pipeline study, and the figures are explicit about the omission on their face.

References

Bibliography

- Shinji Deguchi and William D. Watson. Diffraction in gravitational lensing for compact objects of low mass. *The Astrophysical Journal*, 307:30–37, 1986. doi: 10.1086/164389.
- Daniel J. D’Orazio and Abraham Loeb. Repeated gravitational lensing of gravitational waves in hierarchical black hole triples. *Physical Review D*, 101:083031, 2020. doi: 10.1103/PhysRevD.101.083031.
- Richard A. Isaacson. Gravitational radiation in the limit of high frequency. i. the linear approximation and geometrical optics. *Physical Review*, 166:1263–1271, 1968. doi: 10.1103/PhysRev.166.1263.
- Michele Maggiore. *Gravitational Waves, Volume 1: Theory and Experiments*. Oxford University Press, 2008.
- Carl D. Murray and Stanley F. Dermott. *Solar System Dynamics*. Cambridge University Press, 1999.
- Bohdan Paczynski. Gravitational microlensing by the galactic halo. *The Astrophysical Journal*, 304:1–5, 1986. doi: 10.1086/164140.
- P. C. Peters. Gravitational radiation and the motion of two point masses. *Physical Review*, 136(4B):B1224–B1232, 1964. doi: 10.1103/PhysRev.136.B1224.
- P. C. Peters and J. Mathews. Gravitational radiation from point masses in a keplerian orbit. *Physical Review*, 131(1):435–440, 1963. doi: 10.1103/PhysRev.131.435.
- Philip C. Peters. Index of refraction for scalar, electromagnetic, and gravitational waves in weak gravitational fields. *Physical Review D*, 9:2207–2218, 1974. doi: 10.1103/PhysRevD.9.2207.
- Peter Schneider, Jürgen Ehlers, and Emilio E. Falco. *Gravitational Lenses*. Springer-Verlag, 1992.
- Ryuichi Takahashi and Takashi Nakamura. Wave effects in the gravitational lensing of gravitational waves from chirping binaries. *The Astrophysical Journal*, 595(2):1039–1051, 2003. doi: 10.1086/377430.
- Andrew Ulmer and Jeremy Goodman. Femtolensing: Beyond the semiclassical approximation. *The Astrophysical Journal*, 442:67–75, 1995. doi: 10.1086/175422.